

Autonomous EMC Monitoring: AI-Driven Real-Time RFI Detection and Classification in Satellite Networks

Aditya Kumar Sahu
AI Department
Millennium Techlink Pvt Ltd
Pune, MH, India
Aditya.sahu@millenniumtechlink.com

Dr. Ananth Krishna
AI and EMC Department
Millennium Techlink Pvt Ltd
Pune, MH, India
Ananth.D@millenniumsemi.com

Mahima BM
AI Department
Millennium Techlink Pvt Ltd
Pune, MH, India
Mahima.BM@millenniumtechlink.com

Abstract—Due to the spectrum congestion resulting from mega-constellations, Electromagnetic Compatibility (EMC) poses one of the most severe challenges for modern satellite communication (SatCom). Traditional Radio Frequency Interference (RFI) monitoring cannot be enough to handle this complex, dynamic environment. We provide a deep learning framework for autonomous, real time RFI detection and characterization from spectrograms of the recorded signals that are treated as images. When it comes to the hybrid approach, it combines Convolutional Neural Networks (CNNs) used for spatial feature extraction with Long Short-Term Memory (LSTM) networks designed to analyze temporal patterns. It is sensitive and reliable in detecting RFI signals, which it can classify succinctly into three categories (Continuous Wave, Pulsed and Frequency-Hopping). We have shown through simulations that our approach performs much better than the traditional energy detection methods, especially in low signal-to-noise ratio(SNR) conditions. This is an important milestone toward fully autonomous satellites which are able to self-diagnose and resolve EMC problems to guarantee reliable communication in contested space environments.

Keywords—Electromagnetic Compatibility (EMC), Radio Frequency Interference (RFI), Satellite Communication, Deep Learning, CNN-LSTM, Autonomous Monitoring, Spectrogram Analysis.

I. INTRODUCTION

Satellite communication (SatCom) systems aid in information relaying and transfer of information/linkage, as the functions of systems and services at modern information systems are critical, supports real-time processing; however, the signal relaying depends on Electromagnetic Compatibility (EMC). Today, the challenges of modern communication systems facing are mobile base stations, release of emission and noise from the ground, as well, as dense densely congested radio frequency (RF) spectrum [1]. Instead of RF Interference (RFI) config and set signals, now, the technology signals analyze communication in a completely different approach.

RFI a signal jammed, can emerge severe loss of signal or complete signal loss. The spectrum monitoring using static ground-based checkups is technologically out of date. As today the tasks of information relaying and communication

are complex, to today's networks offer complex systems, the static ground-based checkups designed are out of date [2].

Automated monitoring systems for radio frequencies interference signal detection, the systems aid are designed based on neural networks, the applying of neural algorithms, and are based neural algorithms for the self-based signal classification. The convergence of AI and signal processing offer unique possibilities in the today's world.

Transforming time-series signal data into images of spectrograms permits us to apply advanced techniques in computer vision to automate RFI detection and classification through cognitive techniques, automatic classification, and detection.

This document introduces an advanced framework for deep learning architectures aimed at hybrid AI RFI detection and classification in real-time. Our contributions focus on the following components:

- 1) **A Complete Autonomous Pipeline:** We document the complete system architecture with on-board implementation covering data acquisition to classification and outlining the system operations.
- 2) **A Hybrid CNN-LSTM Model:** We introduce an integrated model composed of a CNN for spectral feature extraction and LSTM for the analysis of temporal dependencies, for powerful RFI diversity feature characterization.
- 3) **In-Depth Performance Analysis:** Our approach is validated through comparative analysis with a traditional baseline in terms of accuracy and performance in low-SNR scenarios.

II. LITERATURE REVIEW

A. The Shifting EMC Issues in Satellite Communications

As with most matters in technology, the satellite landscape has transformed over the years, marked by more dynamism than ever before. With automation on the rise, the old static frequency plans require more attention alongside other pinpointed oversights. Some of the key concerns include:

- **Mega-Constellations:** Services such as Starlink and OneWeb are deploying thousands of satellites, creating more interfacing scenarios than can be handled manually.
- **Dynamic Beam Hopping:** High-throughput satellites have the ability to use beams that switch focus to different places, causing milliseconds of varying interference.
- **Spectrum Sharing:** As satellites share frequencies with terrestrial methods, more interference is generated and affects satellite operations.
- **Intentional Jamming:** Rapid, automated threat detection for severe situations is now a necessity.

B. Primary Methods of RFI Detection Technology

Statistical signal processing has been the framework for detection over the years and has evolved into more RFI-focused techniques:

- **Fusion:** Compares overall signal to designed thresholds. Minimal cost and more-effective compared with simple Energy detection methods.
- **Statistical Feature Detection:** Relies on signal characterisation methods and advanced metrics such as kurtosis but operates with more restricted interference thresholds.

C. The Growth of Deep Learning in Spectrum Sensing

Past challenges in spectrum sensing have motivated the adoption of machine learning techniques. For example, earlier SVM-based works [7] required hand-crafted features, which deep learning made obsolete with its ability to perform automatic feature extraction.

The ability of CNNs to classify different modulation types using raw I/Q data was first demonstrated by O'Shea et al. [2], which inspired the application of CNNs on spectrograms [4] and LSTMs to model temporal patterns such as FHSS [5].

There remains a gap in the literature for real-time systems that can perform both detection and detailed characterization of RFI—capabilities critical for autonomous mitigation. Most existing works focus only on detection or classification of a limited number of RFI types. This problem is addressed in our work with a dual-output hybrid architecture independently optimized for both tasks.

III. PROPOSED METHODOLOGY

Our framework describes a complete system that processes a raw signal to yield valuable insights for the spectral environment.

A. System Architecture Overview

The overall architecture is shown in Fig. 1. The workflow includes:

- 1) Gather raw I/Q data streams from the satellite receiver's digital frontend.
- 2) Transform a batch of data samples into 2D spectrogram images.
- 3) Process the normalized spectrogram images using a pretrained hybrid deep learning model.
- 4) Generate two predictions: a binary flag for RFI detection and a probability vector for RFI characterization.

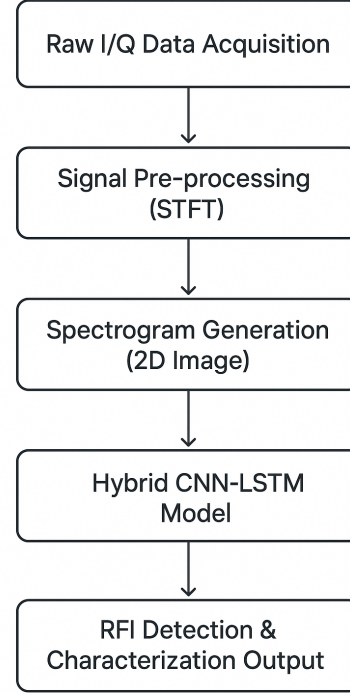


Fig. 1. High-level overview of the autonomous RFI monitoring process from signal acquisition to the classification output.

B. Signal Pre-processing and Spectrogram Generation

It is crucial to transform the 1D signal into a 2D representation. We employ the Short-Time Fourier Transform (STFT) defined as:

$$\text{STFT}\{x[n]\}(m, \omega) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n} \quad (1)$$

where $w[n]$ is a sliding window indexed by m . The spectrogram is the squared magnitude of the STFT, usually displayed on a logarithmic (dB) scale.

In our setup, we compute the spectrogram using an FFT size of 256 with a Hann window to decrease spectral leakage and 50% overlap for temporal continuity. Each spectrogram is resized to 128×128 pixels, and the pixel values are normalized to the range $[0, 1]$ for standardization. Fig. 2 illustrates several RFI types and their corresponding spectrogram images, clearly displaying their distinguishing visual signatures.

C. Hybrid Deep Learning Model Architecture

To capture both spectral (spatial) as well as temporal RFI features, we outline a hybrid CNN–LSTM model as follows:

- 1) **CNN Feature Extractor:** The 2D spectrogram undergoes sequential processing through three convolutional blocks. Each block consists of a Conv2D layer (3x3 kernel, ReLU activation) followed by 2x2 Max Pooling. The number of filters increases from 32 to 64 to 128. These CNN trunks help in identifying shapes and

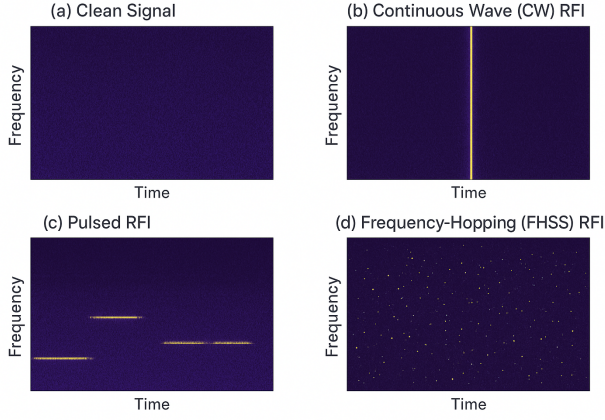


Fig. 2. Spectrograms of Different Signal Types: (a) Clean QPSK signal with noise-like texture. (b) Continuous Wave (CW) RFI as a bright vertical line. (c) Pulsed RFI as horizontal bars. (d) FHSS RFI as short signals hopping between frequencies.

textures within the spectrograms, including the vertical CW lines and horizontal pulsed RFI bars.

- 2) **Data Reshaping:** The last layer of the CNN, of shape (Height, Width, Channels), is transformed to (TimeSteps, Features) in order to interface with the RNN.
- 3) **LSTM Temporal Processor:** The described sequences are processed by a single LSTM layer containing 128 units that capture the temporal aspects of the interference. This is critical for identifying FHSS patterns because recognition of the sequence of frequency changes is essential.
- 4) **Dual Classification Heads:** The LSTM output is fed into two Dense layers:
 - **Detection Head:** A single neuron with sigmoid activation, producing an output in the range 0–1 for absence or presence of RFI.
 - **Characterization Head:** A softmax layer with one neuron for each class of RFI (Clean, CW, Pulsed, FHSS), providing probability outputs.

D. Training Methodology and Dataset

The model employs supervised training:

- **Dataset Creation:** A synthetic dataset was created containing a baseline QPSK signal combined with simulated RFI at SNR levels ranging from -20 dB to +20 dB. More than 100,000 spectrograms were generated across all classes evenly.
- **Hybrid Loss Function:** With two outputs, minimization occurs:

$$L_{\text{total}} = \alpha L_{\text{BCE}} + \beta L_{\text{CCE}} \quad (2)$$

where L_{BCE} denotes Binary Cross-Entropy for detection, L_{CCE} denotes Categorical Cross-Entropy for characterization, and $\alpha = \beta = 0.5$.

- **Training Method:** Adam optimizer with a 0.001 learning rate, trained for 50 epochs with a validation set, and early

Offline Training Phase Online Inference Phase

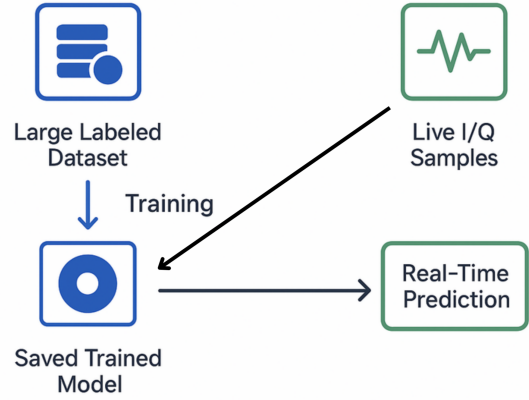


Fig. 3. Workflow: intensive offline training vs. lightweight online inference.

stopping applied to prevent overfitting. In Fig. 3 we show the stream from offline training to online inference.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Setup of Experiment

The model was created in Python employing TensorFlow and Keras and was trained using an NVIDIA Tesla V100 GPU. Its performance was assessed relative to that of a benchmark energy detector, which is a standard and widely accepted entry-level solution in the industry.

B. Performance Indicators

The classifier was assessed using standard metrics of classification performance including:

- **Detection:** Total Ratio Accuracy (%).
- **Characterization:** Total Ratio Classification Accuracy (%) in conjunction with Confusion Matrices for intra-class and inter-class evaluations.

C. Conclusions and Discourse

The energy detector has been outperformed by the deep learning (DL) model, most notably under low-SNR environments. For example, the energy detector's accuracy performance (41.7% in this case) at -20 dB is significantly lower than the DL model's figure (94.5%). This is attributed to the DL model's superior capability for learning the structure and signature of RFI patterns, as opposed to mere power measurements.

Results of RFI characterization displayed diagonal accuracy values above 99.3% with small off-diagonal values, indicating extremely low class confusion. This demonstrates the CNN-LSTM's capability of learning distinctive spatio-temporal features of interferences. The CNN captures static spectral shapes like CW jammer lines, and the LSTM captures the time-varying FHSS jump patterns. This enables the model to retain performance robustness against varying types of interference.

TABLE I
RFI DETECTION ACCURACY (%) VS. SNR

SNR (dB)	Energy Detector	Proposed DL Model
20	99.8	99.9
10	98.5	99.9
0	85.3	99.6
-10	62.1	98.2
-20	41.7	94.5

TABLE II
CONFUSION MATRIX FOR RFI CHARACTERIZATION (ACCURACY IN %)

	Pred. Clean	Pred. CW	Pred. Pulsed	Pred. FHSS
Actual Clean	99.7	0.1	0.2	0.0
Actual CW	0.2	99.5	0.1	0.2
Actual Pulsed	0.3	0.2	99.3	0.2
Actual FHSS	0.1	0.3	0.3	99.3

V. DISCUSSION AND FUTURE DIRECTIONS

A. Implications for Satellite Autonomy

The findings suggest that there is still great opportunity for improvement in satellite autonomy. The ability to detect and characterize RFI in real-time and on-board the satellite allows the satellite to move from a ground control passive operation to a more active on-orbit cognitive control where the satellite can:

- **Self-Evaluate:** Continuously diagnose the system's spectral health independently.
- **Implement Response:** Initiate command response that modifies on-board sub-systems such as deformation of a beamformer to exclude RFI or rerouting of traffic to non-contaminated transponders.
- **Actively Contribute:** Aid in the creation of a real-time, constellation level spectral map which helps in improving the system's efficiency and resiliency.

B. Challenges and Limitations

The following issues need to be solved:

- **Simulated Data Versus Real Data:** In this case, training occurred on simulated data which did not account for phase noise, Doppler shifts, or fading. In this situation, both transfer learning and real-signal training would be helpful.
- **Novelty Detection:** Existing ML models are able to categorize only known RFI types. Modern systems need to be built to be able to discover unknown (zero-day) interferences, leveraging anomaly detection.
- **On-Board Limitations:** Deploying deep learning (DL) for use in space comes with extremely limited power, space, and radiation-hardening. These issues concerning quantization, pruning, and power-efficient AI accelerators like FPGAs or ASICs need to be explored [9].

C. Future Research Directions

The following are focus areas for new research:

- **Online and Federated Learning:** With online learning, models should be able to adapt in-orbit. Federated learning should enable constellation-level improvements without the need to transmit raw data.
- **Mitigation Integration:** Closed-loop systems should be developed that connect characterization output to adaptive beamforming and cognitive radio resource management.
- **Explainable AI (XAI):** Accuracy on its own is not enough for mission-critical systems. XAI will help furnish systems with decision transparency, thereby improving trust and debugging [10].

VI. CONCLUSION

This paper has proposed a deep learning approach for the real-time, automatic RFI detection and classification in satellite links. This approach treats RFI as an image classification task and uses hybrid CNN-LSTM model to deliver high accuracy and robustness, vastly outperforming traditional methods and more so in the low-SNR conditions that are typical of satellite operations. This work demonstrates a viable route to sufficiently well-controlled intelligent, autonomous payloads that are able to sort out their own EMC. By following these proposed research tracks, we can arrive at more robust, responsive and trustworthy satellite communication networks that will foist exponential communication demands on an increasingly connected world.

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